Chapter 1

Introduction to Clinical Practice, Research, and Pharmacy Education

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Learning Objectives:

Objective 1.1 Describe salient features of clinical practice of pharmacists.
Objective 1.2 Address relevant concepts in pharmacy education leading to clinical practice.
Objective 1.3 Outline how research in pharmacy is improving patient outcomes.

OBJECTIVE 1.1. DESCRIBE SALIENT FEATURES OF CLINICAL PRACTICE OF PHARMACISTS

According to the *Oxford Dictionary of English,* the term "clinical" is used in various medical fields to mean that it relates to the treatment of actual patients. Pharmacists, regardless of where they are working, are clinical staff and healthcare professionals. In 2005, the American College of Clinical Pharmacy (ACCP) defined clinical pharmacy as

> *a health science discipline in which pharmacists provide patient care that optimizes medication therapy and promotes health, wellness, and disease prevention.*[1]

The practice of clinical pharmacy embraces the philosophy of pharmaceutical care[2]; it blends a caring orientation with specialized therapeutic knowledge, experience, and judgment to ensure optimal patient outcomes. All direct patient care activities such as patient counseling, medication reconciliation, medication therapy assessment, and any other patient care service are clinical services at the bedside or otherwise (Figs. 1.1 and 1.2). Practice-based research must contribute to the generation of new knowledge that improves practice and the quality of life of patients.[2]

Significant developments in the pharmacy profession have occurred since the introduction of the concept "pharmaceutical care" in the early 1990s.[2] Pharmacists provide a wide range of clinical services[3] irrespective of the setting in which they work.[4] Advising a patient to take an irritant drug with plenty of water will improve quality of life if the patient is adherent to the advice. The clinical expertise of a pharmacist helps to judge if the patient would use medicines appropriately. Tailoring advice or other supporting services to improve adherence of patients to therapy and actions to be taken to prevent adverse outcomes are clinical services. In some countries, pharmacists have also been granted independent or supplementary prescribing rights. Administration of certain vaccines is another such clinical service provided by pharmacists in many jurisdictions. Pharmacists have a long history of managing minor ailments with nonprescription products or just with lifestyle advice. All these activities involve clinical skills.

Pharmacy education has evolved to address the necessary clinical pharmacy competencies. Bachelors of pharmacy programs in many countries have less clinical education, though its importance is increasing. To develop expertise in clinical pharmacy means obtaining a specialized master's program. Although bachelors make pharmacists, masters in clinical pharmacy make clinical pharmacists. In the United States, Doctor of Pharmacy (PharmD) is the only program that makes pharmacists. Pharmacists are considered to have clinical expertise in community pharmacy, hospital pharmacy, or

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FIGURE 1.1 Clinical service at the bedside. *From Bob Nichols (photographer)* https://commons.wikimedia.org/wiki/File:20130306-OC-RBN-3902_(8575102773).jpg.



FIGURE 1.2 Patient counseling at the pharmacy. *From Rhoda Baer (photographer)* https://commons.wikimedia.org/wiki/File:Woman_consults_with_pharmacist.jpg.

other clinical pharmacy settings. Being a clinical pharmacist in a specialized area may require additional certification or residency.[5]

Community-based pharmacy practitioners are highly accessible healthcare professionals. Ambulatory patients and other consumers benefit from pharmacists' clinical care and supply of healthcare products. Timely access to quality drugs is important for the patient. Providing the right drug at the right time is part of a pharmacist's duty of care. Institutional pharmacists usually deliver more complex clinical services in addition to their nonclinical roles. Pharmacists have always responded clinically to people's health needs, improving their access to quality drugs and providing advice about drugs. Pharmacists with full-time clinical responsibilities provide their services mostly at the bedside.[6]

Pharmacists are well educated to interact with patients concerning their diseases and medication. Pharmacists possess in-depth knowledge of medications that integrates with a foundational understanding of the biomedical, pharmaceutical,

sociobehavioral, and clinical sciences. To achieve the desired therapeutic goals, the pharmacist applies evidence-based therapeutic guidelines, evolving sciences, emerging technologies, and relevant legal, ethical, social, cultural, economic, and professional principles.

Accordingly, clinical pharmacists assume responsibility and accountability for managing medication therapy in direct patient care settings, whether practicing independently or in consultation/collaboration with other healthcare professionals. With increased patient care activities, it is likely that more pharmacists are involved in medication management and assume some level of prescribing or advising authority, as has already occurred in some countries.[7] The incidence of errors in pharmacist-written discharge medication orders in a study from the United Kingdom was found to be low.[8]

The clinical practice of pharmacists is evidence based. Pharmacists generate, disseminate, and apply new knowledge that contributes to improved health and quality of life. Pharmacists also supply evidence-based therapeutic advice to other healthcare professionals.[1]

To discuss as a model, the American Association of Colleges of Pharmacy (AACP) consistently tries to standardize the educational outcomes of pharmacy programs in association with other pharmacy organizations. Three important documents that explain the educational outcomes of pharmacy undergraduate programs in the United States are as follows:

- Center for the Advancement of Pharmacy Education 2013 Educational Outcomes (CAPE 2013)[9]
- Pharmacists' Patient Care Process (PPCP)[10]
- Entrustable Professional Activities (EPAs)[11]

AACP identified EPAs for pharmacy graduates. Core EPAs for New Pharmacy Graduates are discrete, essential activities and tasks that all new pharmacy graduates must be able to perform without direct supervision on entering practice or postgraduate training and operationalize the CAPE Educational Outcomes.[11] The 15 CAPE Educational Outcomes are relevant and consistent with emerging scientific and clinical developments and practitioner roles.[9] Continued development of practice-specific EPAs are necessary to extend the definition of what a pharmacist is and does beyond the new graduate. It also helps to improve the navigation of a clinician from one practice setting to another.[12]

PPCP is a consistent patient-centered care process that pharmacists use in collaboration with other healthcare providers, effective in communication and documentation to optimize patient health and medication outcomes.[10]

Competencies required for a clinical pharmacist are listed by ACCP in the following six domains:

- direct patient care,
- pharmacotherapy knowledge,
- systems-based care and population health,
- communication,
- professionalism, and
- continuing professional development.[13]

The International Pharmaceutical Federation (FIP) has developed pharmaceutical workforce development goals to establish the milestones for impactful global development of pharmacy professionals. The goals categorized under academy, professional development, and systems are used as benchmarks for quality improvement.[14] Many countries have similar guidelines to standardize pharmacy education and practice.

OBJECTIVE 1.2. ADDRESS RELEVANT CONCEPTS IN PHARMACY EDUCATION LEADING TO CLINICAL PRACTICE

Including more clinical components in pharmacy education is an increasing global trend. The traditional undergraduate pharmacy program has been revised in most institutions to include more clinical components. The PharmD program originally had a high focus on clinical sciences and experiential education. Standardization of pharmacy education systematically embeds essential competencies required in a pharmacist.[9,11] A key statement by FIP on pharmacy education is the "Nanjing Statement on Pharmacy and Pharmaceutical Sciences Education."[15] The statement describes the envisioned future for pharmaceutical education needed to enhance professional standards worldwide.[15] Pharmacy workforce development goals by FIP range from academics to early career to advancing further with continuing education.[16]

The number of pharmacy colleges/schools and pharmacy graduates has increased globally and has resulted in an increased need for experiential educational sites to provide clinical training and experienced educators with clinical expertise.

Educational outcomes of a pharmacy program are usually categorized into three generic domains such as knowledge, skill, and competence. The pharmacy workforce is developed with these predefined abilities. Knowledge includes recall

and understanding; skills are cognitive and psychomotor; and competence is autonomy, responsibility, role in context, and self-development in the affective domain.[17]

Many adult learning methods are used in modern clinical pharmacy education curricula. Student centeredness—the importance of students in educational decision making—is now widely recognized; however, in many regions of the world, education remains teacher centered. The role of the teacher is shifting toward a facilitator, who supports the psychological and physical environment of the students to learn. Instead of students being the visitor of teachers' wisdom, the teacher visits the student's intellect and works on it with the student. Student centeredness should come next to patient centeredness. However, patient centeredness in clinical practice occurs faster than student centeredness in education. More importantly, students should accept patient centeredness even in didactic courses. Patients are the ultimate beneficiaries of pharmacy education that drives achievement of required patient care outcome educational competencies.[18]

Many modes are used for the delivery of course content. Blended learning uses a combination of face-to-face and online learning methods. It is also termed "the flipped classroom" as the lecture and supporting information are provided online before the class and assignments are completed in class—the reverse of traditional teaching methods. Blended learning requires the physical presence of both teacher and student, with some element of student control over time, place, path, or pace. Multiple delivery modes give students the freedom to choose their preferred mode of learning for a particular topic. Delivery modes may include lectures, videos, assignments, research, group discussions, and experiential placements, some or all of which may be optional. Blended learning should be planned considering the availability of technology support and training of teachers to be most useful for the students.[19] Learning is more efficient and interesting with the use of multiple modes of educational delivery. Multimodes of delivery help to reinforce learning. Use of prerequisites gives more time for slow learners to master the courses before they progress into more advanced courses.[20] Teachers should be trained well to perform as facilitators of appropriate modes of learning to their students.

Curriculum revisions to improve global health professional education are occurring to include elements such as integrated active learning, mastery learning, and early and extended clinical exposure. Changes in methods of course delivery encourage active learning, communication, and critical thinking. Some of the following approaches are used in pharmacy education to encourage the students to be more involved in their learning:

- Interactive lectures: A common criticism is that lectures are a passive method of learning. However, they are less expensive to deliver and more convenient than modern active learning methods. One method of making lectures more active is by including interactive sessions, preferably by the use of technology.[21] Other than lectures, methods with high faculty/student and student/student interactions provide active learning such as team-based and problem-based learning (PBL).
- Team-based learning (TBL): Psychologist Bruce Tuckman explained that teams have the following phases: forming, storming, norming, performing, and adjourning. In the stage of forming, a facilitator plays a dominant role as the team members have not yet identified their roles. In storming, the members resist working together. In norming, the members make compromises to work together in a team. In performing, the team is more productive. There are no clear boundaries between these phases. On any new task, the team may slip back to storming stage, especially in the presence of conflicts. Adjourning is a phase opposite to forming. The teams could be dissolved or rearranged at this stage. Many feel uncomfortable in the discontinuation of a good group with understandings of each other. The new teams may need to go through all the phases again.[22] TBL requires the student to be prepared before the class. Each individual student completes an assessment, the individual Readiness Assessment Test (iRAT). In the next step, individuals in small teams discuss and answer the same questions, the team Readiness Assessment Test (tRAT), with correct answers given after each attempt to influence a further decision-making process. Faculty should review the answers to questions students struggled with. Finally, the team discusses the relevance of the topic studied to real practice.[23] Peer learning happens in teams. Team dynamics affect the learning process. Identify and resolve the lack of coordination between teams and within team members to improve outcomes. It is important for team performance that students are accountable to themselves and the team. Frequent and timely feedback from the facilitator is important for learning to be directed toward the expected outcomes. Some pharmacy institutions have an integrated, team-based curriculum. TBL is equally, or sometimes more, beneficial than traditional learning methods in academic performance. The conversion of a part of a course or a group of courses in a program to TBL is commonplace in pharmacy education. In many cases, most students find it more satisfying educationally when trained faculty properly implements it. Faculty training is important to organize TBL systematically. However, students may not like the increased time spent in learning and the demand for more self-directed learning, especially when they change from a lecture to an activity session.[24]
- PBL: Students engage in a problem and a series of progressive disclosures (triggers) to identify the gaps in their understanding that need further self-directed learning. Students have chances to generate learning objectives.

Many trained facilitators are required to provide feedback to students for PBL sessions to be effective. PBL, when conducted properly, helps in problem-solving skills, self-directed learning, collaboration skills, and intrinsic motivation of students. Students learn to identify their knowledge gaps and determine methods to learn further. Use of real-life clinical cases equips students with potential practical skills in a given setting. Shifting from a traditional learning method to PBL is a major curriculum change. As in other healthcare professional education, the academic performance of pharmacy students has been found to be higher with PBL compared with traditional learning in some settings/course components.[25]

- Case-based learning (CBL): CBL is common in pharmacy education. Pharmacotherapy books include cases that lead to active learning. CBL is relatively more structured than PBL. The problem to be solved in CBL (guided inquiry) is provided in advance; however, in PBL (open inquiry), students need to determine the problem for themselves.[26]
- Simulation-based learning: Simulation in clinical experiential learning is common. Simulation, using standardized patients, virtual patients, and manikins, helps the student learn without the fear of causing harm to a real patient and decreases the anxiety of the patients who are being treated by students rather than by an experienced clinician. A standardized patient is a real person who is trained to act like a model patient. Such patients, through interactive virtual platforms, increase convenience and reduce costs. The manikin is used widely even in traditional lectures to demonstrate some course content. One of the most common uses of manikins is when learning basic life support in cardiopulmonary resuscitation. Simulation centers may also be called clinical skills laboratories, where pharmacy students can practice their patient physical assessment skills and provide certain pharmaceutical care services such as vaccinations, medication dosing, or dispensing. Virtual patient platforms are available to perform these tasks. Hybrid simulation is a combination of more than one method, e.g., a standardized patient and a task trainer demonstrating a procedure in a manikin. Simulation is an efficient method of developing practical skills if modeled properly. Simulation scenarios are adjustable to tailor the educational needs of a particular user. Simulation is especially useful in critical care to improve skills to manage patients in crisis situations.[27]

Integrated curriculum: Some level of integration of courses is an inherent component of the pharmacy curriculum. Biomedical and pharmaceutical courses provide a basis for learning clinical sciences. Medicinal chemistry, pharmaceutics, and pharmacology are prerequisites for pharmacotherapeutics, and it is integrated with clinical pharmacy services for better application to practice. In an integrated curriculum, these courses are studied together in modules to help students connect sciences better. Studying, e.g., medicinal chemistry, pathophysiology, pharmacology, pharmacotherapy, and the related pharmaceutics together provide horizontal integration. Horizontal integration brings together disciplines when studying. Horizontal integration is studying different basic sciences (medicinal chemistry, pharmaceutics, pharmacology, etc.) together as well as studying basic and clinical sciences together. Sometimes, the study of basic and clinical sciences together in a unit or year of study is also called horizontal integration. Basic and clinical sciences could be integrated both horizontally and vertically. Vertical integration has also been applied to spiral curricula. The topics are revisited in increasingly complex ways as the program progress. Grounding clinical sciences in the basic sciences or advanced clinical experience in the early clinical experience are examples of vertical integration. A scholarly pathway and a professional skills pathway across many units of study are examples of vertical integration. Early clinical exposure by introductory pharmacy practice experience and advanced experiential training toward the end of the program are other examples of vertical integration.[28,29] Adoption of horizontal and vertical integration at some level is becoming common in pharmacy education. The institution with an integrated pharmacy curriculum mostly has the core sciences in modules rather than in independent courses.[30]

Experiential learning: Students undergo at least two phases of their program: a didactic phase for the foundation courses and an experiential learning phase incorporating the clinical pharmacy education. The transition from didactic to experiential is not abrupt. In the foundation courses of biomedical and pharmaceutical sciences, students have some practical laboratory work. As well, there are didactic sessions in clinical pharmacy in the later years of study. Clinical experiential learning is often preceded by simulated learning that uses virtual patients, manikins, and role play. All these active learning and experiential trainings make students preceptor ready, team ready, and practice ready.

- Students who have sufficient knowledge, skills, and behavior to work in direct patient care areas are preceptor ready. Not only do the interns progress to practice independently, they are also able to precept junior students. This progress happens with peer mentoring of junior students within the pharmacy curriculum. Interns in advanced training mentor junior students in early practice exposure.
- Healthcare practice is teamwork. Abilities to work in teams improve the effectiveness of professionals. Interprofessional education and group-based active learning methods prepare students to be team ready.
- Practice readiness happens with experience. The graduates should be able to practice without supervision or minimal supervision.

The gradual transition from didactic learning methods to simulation to experiential learning results in less harm to patients. Some level of early experiential learning in real patient settings starts with introductory pharmacy practice experiences or clinical placements. Early experiential educational sessions in the pharmacy program help to relate what they are learning further to real practice settings. On completion of all didactic courses, students start their full-time clinical experiential learning, advanced experiential training, or internship or Advanced Pharmacy Experience. Advanced training makes the student fit for practice immediately following graduation. A junior practitioner is not exempted from practice competencies and is considered equally as responsible as an experienced practitioner. Experiential education bridges the transition from student to practitioner. Site capacity, preceptor development, and quality of training and assessment are common concerns faced in experiential education.[31]

Objective Structured Clinical Examination (OSCE): As the teaching of clinical practice in a skills laboratory and in experiential learning are significant curricular components, clinical performance examination is essential to assess clinical skills. Providing clinical services requires advanced clinical and communication skills. OSCEs are therefore becoming widely used to assess students' clinical and communication skills. OSCE has multiple stations that direct students to perform certain tasks to test their specific abilities. Critical thinking and problem-solving skills developed through active learning are best assessed using performance in clinical situations using OSCEs.[32,33]

Interprofessional education: Interprofessional education prepares health professionals to be team ready. Collaborative practice demands a high level of communication between different healthcare professionals. Interprofessional education is used to improve collaborative practice and interprofessional communication. Interactions of students with students of different healthcare professions provide a clear understanding of their roles and responsibilities in practice. Some pharmacy institutions have introduced departments or divisions for promoting interprofessional education.[6] The Inter-professional Education Collaborative (IPEC) was established in 2009 by six health professions with the AACP as a founding member. IPEC developed four core competencies and subcompetencies based on the concepts of patient and family centered and community and population oriented. The revised core competencies in 2016 are as follows:

- Work with other healthcare professionals to maintain a climate of mutual respect and shared values (values/ethics for interprofessional practice).
- Use the knowledge of one's role and those of other professions to assess and address the healthcare needs of patients appropriately and to promote and advance the health of populations (roles/responsibilities).
- Communicate with patients, families, communities, and professionals in health and other fields responsively and responsibly that supports a team approach to the promotion and maintenance of health and the prevention and treatment of disease (interprofessional communication).
- Apply relationship-building values and the principles of team dynamics to perform effectively in different team roles to plan, deliver, and evaluate patient/population-centered care and population health programs and policies that are safe, timely, efficient, effective, and equitable (teams and teamwork).[34]

OBJECTIVE 1.3. OUTLINE HOW RESEARCH IN PHARMACY IS IMPROVING PATIENT OUTCOMES

The practice of pharmacists is largely evidence based. Many pharmacists perform practice-based research. Pharmaceutical industries are highly research based. Pharmacy education focuses on preparing pharmacists and pharmaceutical scientists with research competencies. Pharmacy professionals who spend more of their time in research are pharmaceutical scientists. As per the FIP, "a pharmaceutical scientist is a qualified expert in aspects of the science and technology of medical products." It includes, but is not limited to, the discovery, development, manufacture, regulation, and utilization of medical products—embracing how drugs work, how safe and effective products are brought to the market, their impact on the body, and their effect on the prevention and treatment of disease.[35]

Pharmacists and pharmaceutical scientists perform research at different levels and types: practice-based research (action research), laboratory research, and clinical trials. Drug utilization research by pharmacists influences changes in drug use. Increased use of generic drugs and other rational approaches in drug use are examples of outcomes of pharmacy research. For example, the factors that could be studied for generic drug utilization are (1) patient-related factors, (2) formulary management or cost containment, (3) healthcare policies, (4) promotional activities, (5) educational initiatives, (6) technology, and (7) physician-related factors.[36] Pharmaceutical scientists participate in the innovation of new health technology.

Pharmaceutical scientists commonly perform research at pharmaceutical industries, universities, and governmental and nongovernmental agencies. Pharmaceutical industries produce drugs after comprehensive research. New drug development

research for a product lasts longer than a decade, including many studies. Research continues for the marketed product on a larger population by the manufacturer and other agencies and healthcare professionals. Independent research adds value to existing information, and the drug may be withdrawn from the market if new findings are questioning or discouraging its use. Secondary research by combining clinical trials makes more reliable data through systematic reviews/meta-analysis. The sequence of generating and validating information through extensive research about drugs results in an abundant body of knowledge. Easy access, especially online access to such information and tools to filter information, provides clinicians with evidence to support practice. Clinical guidelines are considered on top of the evidence level as they are sets of practical recommendations reflecting the availability of primary and secondary research data.

New drugs and devices bring cures for diseases that were previously incurable and promote innovation in drug delivery to enhance effectiveness and safety. Newer drugs/devices provide a more patient-friendly treatment for conditions otherwise difficult to manage without serious safety concerns, e.g., complete oral regimen for hepatitis C.[37] Identifying responsible genes for diseases helps in targeting specific sites with specific drugs called personalized medicine. Personalized medicine is defined by the European Union as "the right patient with the right drug at the right dose at the right time." Pharmacogenomics is a part of personalized medicine. It is "the study of variations of DNA and RNA characteristics related to drug response." For certain drugs, pharmacists use pharmacogenomics information for ensuring effectiveness and safety in individual patients with affordable and convenient genetic tests. Some pharmacists work as practitioners in precision medicine, especially in cancer care. Pharmacogenomics has allowed for more tailored treatment of a wide range of health problems, including cancer, cardiovascular disease, and HIV/AIDS.[38]

Pharmacy professionals commonly research health outcomes too. Pharmacoeconomics and outcomes research studies affordability and value of pharmacy services and health technology. Economic, humanistic, or clinical outcomes are studied at clinical and societal levels. The evidence generated from these differs with different settings and perspectives.

It is essential for pharmacists to have research skills not just to generate evidence but also to read and understand research literature. Traditionally, pharmacists used their drug information and literature evaluation skills to incorporate best quality evidence into practice. Pharmacists specialized in drug information service are evidence support staff to other healthcare professionals. Pharmacists provide information support through their knowledge and skills in pharmacotherapy. These cognitive services provided by pharmacists need educational self-development investments and access to drug information resources. This improves evidence-based practice, and institutions should invest in such pharmacists and resources. A pharmacy workforce is developed to provide valuable services to patients and other healthcare professionals. Such services provided by pharmacists for the benefit of patients that result in quantifiable improvements in health outcomes and appropriate cost recovery are essential for sustainability.[39]

CONCLUSION

Pharmacists provide clinical services and research to achieve better patient outcomes. Clinical pharmacy is a concept of direct patient care, irrespective of the settings of practice. Preparing graduates for patient-centered, team-delivered clinical roles must be a priority for educators and agencies that oversee pharmacy workforce development. A pharmacist needs to undergo continuing education to enhance practice. A pharmaceutical scientist contributes to producing better drugs. Incorporation of evidence into practice and generation of practice-based knowledge demand research skills in pharmacists too.